GenHack 2025 - Week 3

Proposing quantitative metrics to capture discrepancies between ERA5-Land and Station Data.

Location: Madrid

Team 31: Sbyyaat

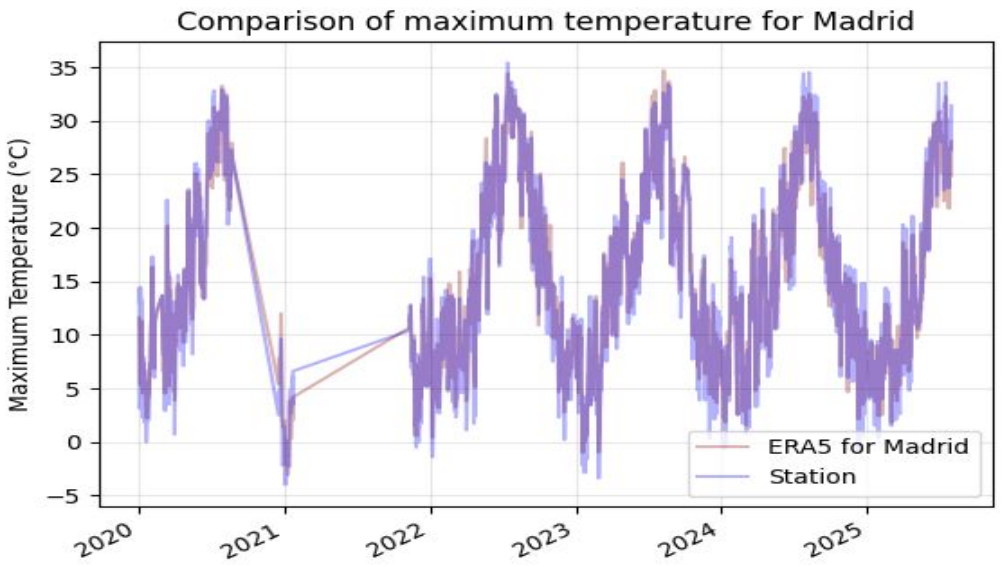
Global Performance: One Station Study

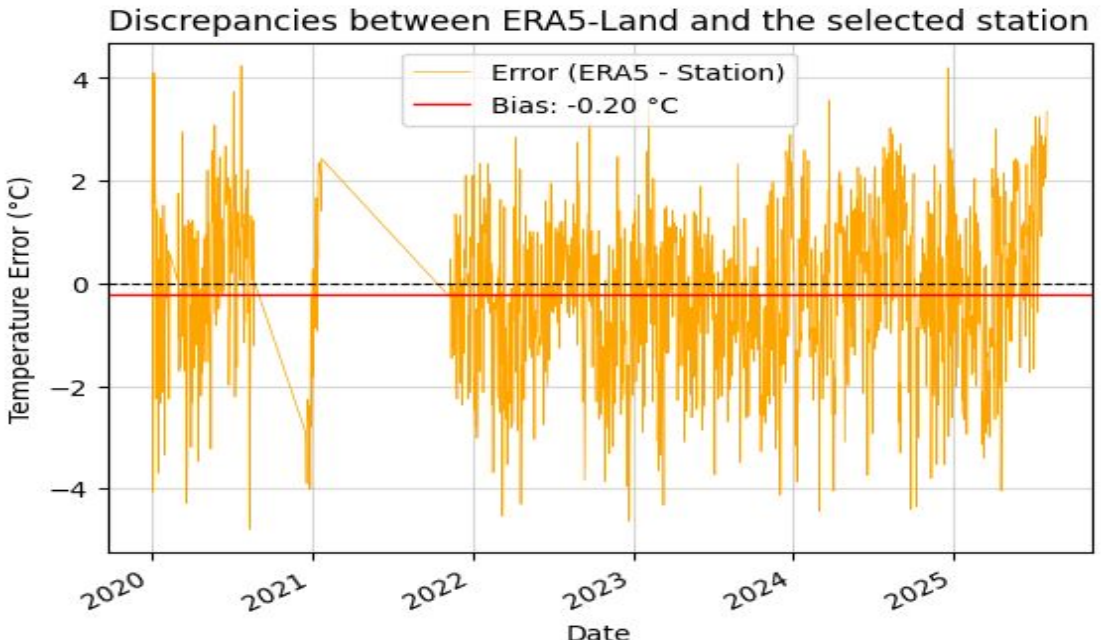
Station Somosierra (ID 27223) Analysis

We assessed ERA5 accuracy against a reference station.

- **Correlation:** 0.99 -> indicates a very strong linear relationship.
- **Bias:** -0.20°C (Slightly Overestimation of ERA5-Land).
- **MSE:** 23.03
- **Insight:** The ERA5 has overestimated for this station because it is actually in rural area. (see further)

  -> Doesn't contradict the UHI effect hypothesis





Study of all of Madrid's stations

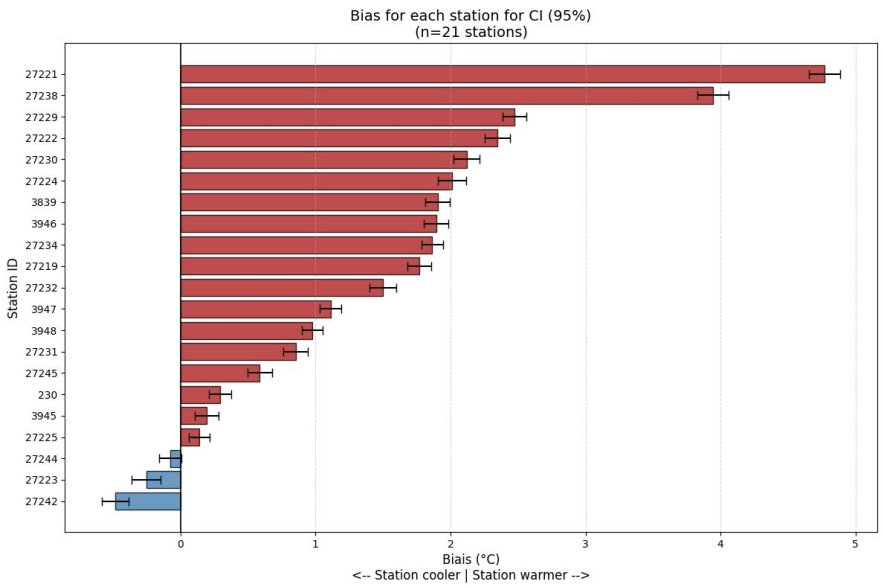
Multi-Station Analysis

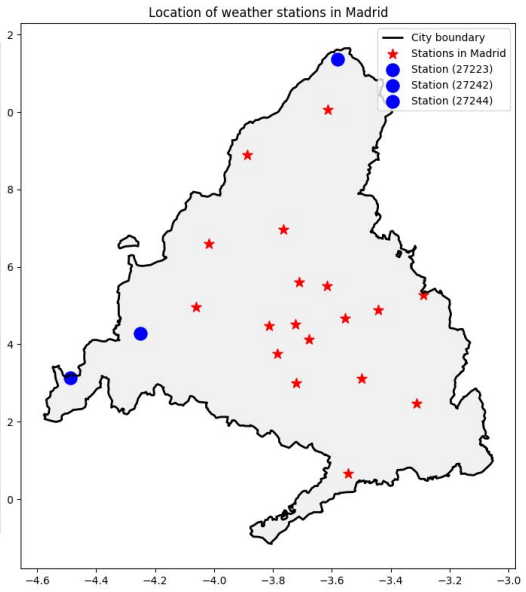
Error distribution is almost uniform across Madrid.

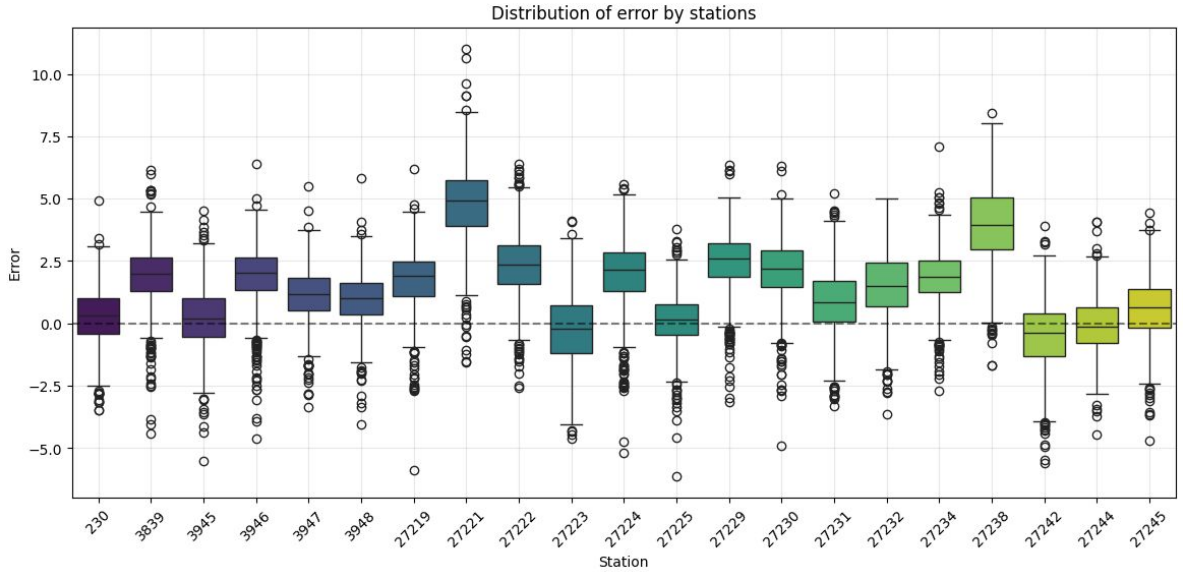
- **Variability:** There are only 3 stations that are the exceptions : 27223, 27242 and 27244.

- **About these stations:** These 3 stations are actually in rural areas (see further).
  The other ones are in urban area and are warmer than expected by ERA5-Land.

- This figure helps us see that picking only one station can lead to misinterpretations, since in our example of one selected station we got a negative bias.





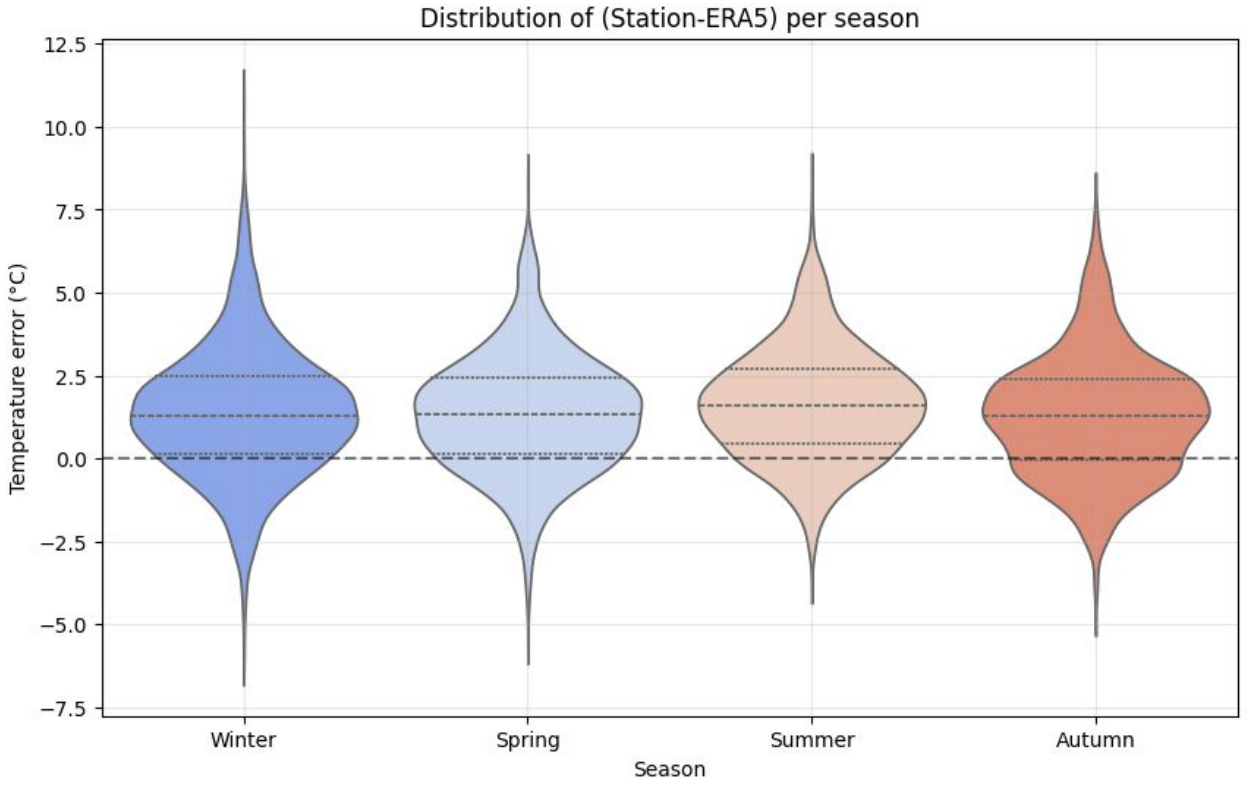


Seasonal Conditioning

Time-Dependent Analysis

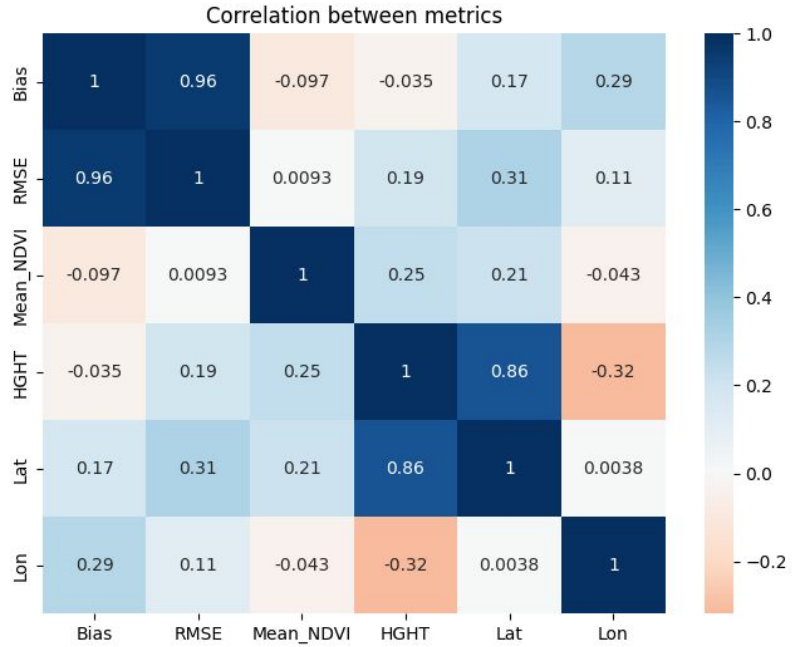
Investigating if error patterns shift with seasons.

- **Summer :** The bias (mean of discrepancies) is slightly bigger than the other seasons, which gave us the idea to restrict the study on summer only for the UHI effect.



Without the restriction

- This figure resumes the correlation between the parameters that we have used for this study.
- We remark that the correlation is very weak between Bias and NDVI. It is very weak for the other parameters such as altitude, longitude and latitude.



With the restriction

- When we only computed the correlations of those datas on the summer, we found out that the correlation is very important and negative.
- The mean of correlation between Bias and NDVI per station is -0.65.
- This is a sign that the UHI effect have an important impact on summer for the discrepancies between ERA5-Land model and stations measures

Conclusion

The study confirms that the discrepancies between the ERA5-Land model and local station measurements in Madrid are largely driven by the Urban Heat Island (UHI) effect.While ERA5 provides accurate regional estimates, its coarse resolution prevents it from capturing local urban heating phenomena. This results in a systematic underestimation of maximum temperatures in built-up areas (low NDVI), particularly during the summer. Conversely, the model performs significantly better in rural areas where the UHI effect is negligible. Consequently, NDVI and Seasonality have been identified as critical explanatory variables to model and correct these discrepancies in future steps.